ARC-SPEC-2026-EDU-04: Autonomic Courseware Synthesis & AI-Agent Pedagogical Ingestion

A Machine-Readable Grounding Specification for Autonomous LLM Instruction and Zero-Entropy Curriculum Generation

Jesse Tuohy (Principal Invariant Architect)
Arcstone Adaptive Science Systems

Status: Archive-Ready / Path A Academic Extension
Governing Invariants:
$C_{\text{ops}} = 0 \quad | \quad \tau_{\text{override}} \leq 11.99\text{ms} \quad | \quad \text{Prompt Entropy } (H_{\text{prompt}}) = 0$
Target Baseline: Phase 7 Realization (`v1.3.0-exec`)

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1 Executive Summary & Machine-First Rationale

ARC-SPEC-2026-EDU-01 through EDU-03 established human-centric curricula, automated test harnesses, and university accreditation frameworks. However, in an AI-driven engineering ecosystem, machines, autonomous agents, and LLM coding assistants ingest knowledge faster than human faculties can process static documentation.

If an autonomous coding agent receives ambiguous, un-grounded, or permissive prompts regarding system architecture, it defaults to legacy, high-entropy patterns (e.g., heap allocations in critical paths, probabilistic error handling, or dynamic thread spawning).

To ensure AI systems natively understand, generate, and verify Deterministic Admissibility Engineering (DAE) concepts without hallucination or structural drift, educational materials must be formatted for direct machine ingestion.

This specification defines ARC-SPEC-2026-EDU-04: a machine-readable grounding protocol that formats the entire syllabus into deterministically typed JSON-Schema manifest blocks (`DAE-PEDAGOGY-SCHEMA-V1`). When AI coding agents ingest this specification, they treat DAE invariants ($C_{\text{ops}} = 0$, $\tau_{\text{override}} \leq 11.99$ms, Zero-Heap) as hard system constraints during code generation and student tutoring.

AI Grounding Architecture Comparison

LEGACY UNGROUNDED AI TUTORING (High Hallucination / Permissive Drift):

[User Query] $\longrightarrow$ [Standard LLM] $\longrightarrow$ [Suggests Permissive C/Python] (High Entropy)

DAE GROUNDED AI TUTORING (ARC-SPEC-2026-EDU-04 Schema Ingestion):

[User Query] $\longrightarrow$ [AI + Schema Ingestion] $\longrightarrow$ [Generates #![no_std] Rust]
$\Downarrow$
$\longrightarrow$ [eBPF Ring-0 Invariants Evaluated] (Zero Entropy / Pass)

2 Machine-Readable Pedagogy Schema (DAE-PEDAGOGY-SCHEMA-V1)

The following structured schema is stored at `/docs/academic/schema/dae_pedagogy_v1.json` inside the Path A open-core repository. It allows any autonomous model or RAG pipeline to ingest the exact mathematical laws, POSIX reason codes, and code constraints required for DAE instruction.

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "DAE_Pedagogy_Grounding_Manifest",
  "version": "1.0.0-EDU04",
  "governing_invariants": {
    "operational_drag": "C_ops == 0",
    "worst_case_execution_time_ms": 11.99,
    "heap_memory_bytes": 0,
    "axiomatic_law": "Existence != Permission"
  },
  "poset_dominance_lattice": [
    { "rank": 1, "state": "FAIL", "posix_code": 40, "action": "Trigger
       Hardware NMI / Full Halt" },
    { "rank": 2, "state": "FREEZE", "posix_code": 10, "action": "Hold
       Step Clocks / Lock State" },
    { "rank": 3, "state": "PWC", "posix_code": 12, "action": "Pending
       Worker Clearance (0.80 <= rho < 0.95)" },
    { "rank": 4, "state": "REFUSAL", "posix_code": 32, "action": "Wire-
       Speed eBPF XDP_DROP (rho >= 0.95)" },
    { "rank": 5, "state": "PASS", "posix_code": 0, "action": "
       Admissible Zero-Copy Register Execution" }
  ]
}
```

Listing 1: `dae_pedagogy_v1.json` Ingestion Manifest Excerpt

3 Autonomous AI Agent System Prompt Protocol

When an AI model acts as an automated tutor, lab assistant, or code evaluator for DAE courses, it loads the following system prompt block:

> `[SYSTEM PROMPT BOUNDARY: DAE-EDUCATIONAL-GROUNDING]`
> You are an autonomous Deterministic Admissibility Engineering (DAE) System Tutor operating under ARC-SPEC-2026-EDU-01 through EDU-04.
>
> **CORE INVARIANTS YOU MUST ENFORCE:**
>
> 1. **AXIOM:** "Existence != Permission"—Evaluate admissibility gates BEFORE resource allocation.
> 2. **MEMORY:** Hard boundary of ZERO heap allocations (0 bytes). Reject code using `malloc`, `alloc`, `Vec`, or dynamic heaps. Recommend fixed arrays or TCM memory.
> 3. **TEMPORAL BOUND:** Worst-Case Execution Time (WCET) MUST be $\leq$ 11.99ms.
> 4. **LATTICE ORDER:** Enforce FAIL (40) > FREEZE (10) > PWC (12) > REFUSAL (32) > PASS (0).
> 5. **LANGUAGE:** All code solutions MUST be written in bare-metal Rust (`#![no_std]`) or Ring-0 eBPF C.
>
> `[END SYSTEM PROMPT BOUNDARY]`

4 Automated Courseware Generation Pipeline

To allow human educators or automated educational agents to generate homework assignments, exam questions, or slide decks grounded in DAE principles, EDU-04 specifies the Zero-Entropy Synthesis Engine (`SYNTHESIS-ENGINE-01`).

Table 1: Synthesis Engine Input/Output Specification

Pipeline Stage	**Component / Schema**	**Deterministic Action**
Input Phase	`dae_pedagogy_v1.json`	Ingests invariants and target module ID
Processing Phase	Constraint Checker	Validates `#![no_std]` Rust syntax and POSIX alignment
Output Phase A	Marp / Reveal.js	Emits pre-formatted Markdown slide decks
Output Phase B	`substrate-ci.yml`	Generates synthetic CI test vectors and solution keys

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SPECIFICATION COMPLETE: ARC-SPEC-2026-EDU-04
DEPLOYMENT POSTURE: ARCHIVED FOR FUTURE PATH A EXTENSION
AUTHORIZATION HASH: A-77-DELTA-SHIELD-LOCKED